

Examiner  
only

3. Carbon-12 ( $^{12}_6\text{C}$ ) and carbon-14 ( $^{14}_6\text{C}$ ) are isotopes.

(a) Explain, in terms of the **numbers of particles**, why carbon-14 has an unstable nucleus and carbon-12 has a stable nucleus. [3]

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(b) An unstable nucleus becomes more stable following the emission of radiation. **Complete the table below.** [4]

| Type of radiation | Symbol       | Description               |
|-------------------|--------------|---------------------------|
| gamma             | .....        | Electromagnetic (em) wave |
| alpha             | $^4_2\alpha$ | .....<br>.....            |
| beta              | .....        | .....<br>.....            |

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